

Mikołaj Karkowski

Full-stack .NET Engineer

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PROFILE

Full-stack .NET engineer with 7+ years of commercial experience building web and desktop products for complex, data-intensive domains. I turn specialized requirements into maintainable software—from real-time Blazor interfaces and ASP.NET Core services to WPF clients and applied AI systems. I value clear boundaries, strong typing, practical automation and close collaboration with users.

CORE SKILLS

- Languages & runtime
- C#, .NET 10, T-SQL
- Web
- ASP.NET Core, Blazor WebAssembly, Angular, REST APIs, SignalR
- Desktop
- WPF, MVVM, .NET MAUI Hybrid
- Data
- EF Core, PostgreSQL, SQL Server, SQLite
- AI
- Microsoft.Extensions.AI, OpenRouter, GitHub Models, Ollama, YOLO
- Engineering
- DDD, CQRS, modular monoliths, Docker, .NET Aspire, OpenTelemetry, GitHub Actions
- Testing
- xUnit, NUnit, Moq, WebApplicationFactory

EDUCATION

- Master of Computer Science
- 2021–2022
- Silesian University of Technology
- Bachelor of Automation and Robotics
- 2016–2021
- Silesian University of Technology

LANGUAGES

- Polish — native
- English — C1

INTERESTS

- Photography, hiking, travel

EXPERIENCE

- Software Developer
- Apr 2019 — present
- Graw / Goldschmidt Group
- Design, develop and maintain web and desktop .NET products for railway measurement, inspection and navigation systems. Collaborate directly with clients to shape requirements and deliver effective solutions.
- Built Blazor WebAssembly and .NET MAUI Hybrid interfaces presenting live camera feeds, laser measurements and equipment status with role-based UI access.
 - Created WPF tools for visualizing measurements, reviewing rail defects, displaying equipment diagnostics and working with video-backed image data.
 - Developed an AI-assisted train-navigation algorithm combining video, computer vision, rail infrastructure and operational data as an alternative where GPS is unreliable.
 - Built a rail-object detection tool and trained YOLO models for finding damage on tracks and sleepers.
- Engineering Intern
- Apr 2018 — Jun 2018
- Bobson
- Programmed Arduino boards, tested CAN networks and operated milling equipment.
- Junior Software Developer Intern
- Apr 2015 — Jun 2015
- QNT Systems
- Developed a responsive web application using Bootstrap.

SELECTED PROJECTS

- FinanceManager
- GitHub
- Open-source personal finance platform
- Full-stack .NET 10 product for managing cash, stocks and bonds with analytics, live market data, CSV workflows and AI-generated insights. Built as a layered modular monolith with a pure domain layer, typed HTTP clients, background channels, SignalR, JWT authentication and OpenTelemetry.
- Kotlet
- GitHub
- Connected home-kitchen workspace
- Recipe, pantry, meal-planning and shopping platform with a shared ingredient catalogue. Includes 28-day planning, nutrition and cost calculations, public recipe sharing, image workflows and AI-assisted imports from video sources.